

Transparency in Debt Crisis ^{*}

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Abstract

We study the problem of a sovereign choosing whether to disclose information to international lenders in an [Eaton and Gersovitz \(1981\)](#) environment. The government faces a trade-off: full disclosure ensures that debt is sold at high prices in good times, but hampers new debt issuance in bad times. Conversely, non-disclosure creates an insurance opportunity through adverse selection. Under mild conditions, the unique equilibrium under no-disclosure is a pooling equilibrium that allows the sovereign to take more debt in bad times at the cost of worse prices obtained in good times. We characterize the sovereign's optimal choice of information disclosure and show that non-disclosure is preferred when deadweight losses from defaulting are small. We relate this mechanism to the 1994-1995 Mexican Peso Crisis, when information about international reserves was temporarily withheld. Using IMF data, we link the rise in the number of conditionalities attached to IMF programs to an increase in the costs of default, which in our model shifts the optimal policy toward full disclosure.

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1 Introduction

Lender-borrower relationships are inherently shaped by informational asymmetries. Sovereign debt is no exception: governments typically possess better information about their repayment capacity and policy intentions than their creditors. This information gap creates scope for strategic behavior, as countries decide how much to reveal about their fundamentals to influence market perceptions. In international credit markets, where enforcement is limited and beliefs determine borrowing conditions, the management of information becomes a central dimension of sovereign policy. This paper examines the circumstances under which a sovereign optimally withholds information from lenders and how disclosure rules shape borrowing outcomes.

Mexico's 1994-95 Peso Crisis provides a natural setting. Throughout 1994, even as pressures intensified, the Mexican government maintained its long-standing practice of releasing reserve data only three times a year (March, October, and November) and did not deviate from this practice until after the crisis. Because reserves were used both to service dollar debt and to defend the peso, their level directly signaled repayment capacity. The left panel of Figure (1) shows how reserves relative to short-term dollar debt declined between official releases, illustrating the amount of information accumulated between disclosure dates. The right panel compares contemporaneous press statements with the values later disclosed: points referring to past reserve levels lie close to the 45-degree line, whereas current statements (nowcasts) lie well above it, revealing systematic overestimation. Taken together, these patterns indicate that, before 1995, Mexico followed a sparse and delayed reporting regime that left lenders largely uninformed about the rapid depletion of reserves.

Our primary goal is to describe the trade-offs faced by a government when deciding whether to disclose information to lenders about a privately known state of the world, and to characterize the conditions under which not disclosing this information is preferred. To that end, we analyze a tractable two-period sovereign debt model, following [Eaton and Gersovitz \(1981\)](#), in an incomplete-information setting where the government commits ex ante to either fully disclose its repayment capacity or withhold it. We state parametric assumptions under which there is no separating equilibrium in the model. We also derive conditions under which the pooling equilibrium features cross-subsidization: although all types of sovereigns borrow the same amount, only a subset of them will repay, a fact that is incorporated into bond prices. The repaying sovereigns then bear the costs of being pooled with a defaulter due to the adverse selection problem. Before know-

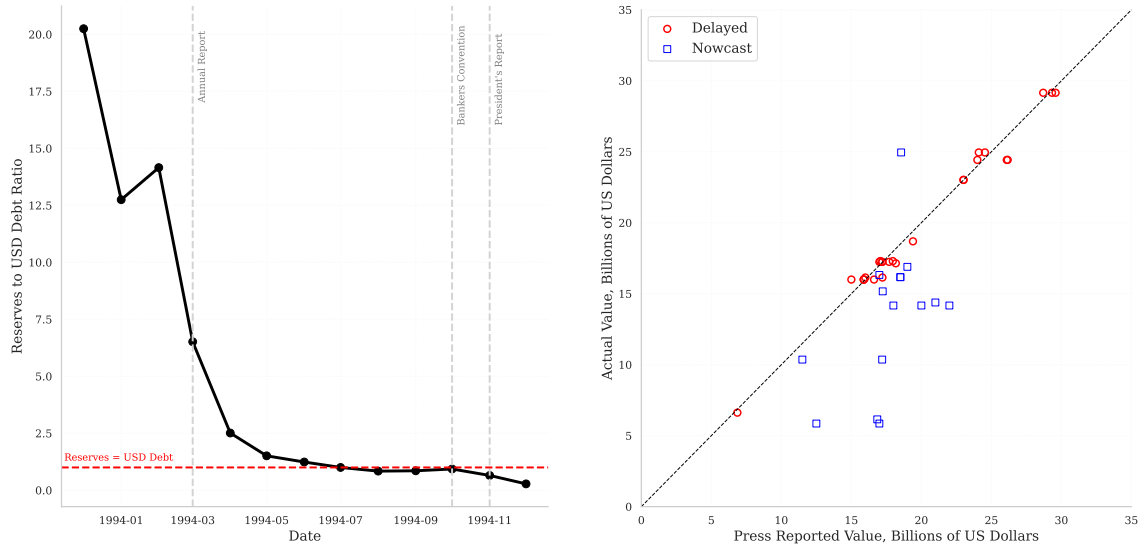


Figure 1: Panel A (left) shows the ratio of foreign exchange reserves to short-term dollar debt in 1994. Vertical dashed lines mark the dates of official reserve releases: the Annual Report (March), the Bankers Convention (October), and the Presidents Report (November). Panel B (right) compares contemporaneous statements about reserve levels with the values later disclosed; deviations from the 45-degree line indicate systematic overestimation before 1995.

ing its type, the sovereign can then obtain insurance against a low repayment capacity by choosing not to disclose information to lenders. We show that this insurance leads the sovereign to choose the pooling outcome over the lottery-like payoff from full disclosure for a larger set of lenders' beliefs when deadweight losses from default are small. We use data from [Kentikelenis et al. \(2016\)](#), and [Akerman et al. \(2022\)](#) to proxy default costs by the number of conditions imposed by the IMF on its deals. We document an increase in the number of conditions per loan after 1995, aligning with the government's decision to move to a higher frequency, full disclosure of reserves.

We endow the sovereign with the ability to commit to an ex-ante chosen information structure, as in [Kamenica and Gentzkow \(2011\)](#). This assumption makes the choice of information structure non-strategic in our environment: the disclosure rule itself does not reveal additional information about the sovereign's type. Lenders take the rule as given and set the price of debt based on the beliefs it induces. The sovereign chooses the rule to shape those beliefs and commits to it. The motivation is empirical: Mexico did not change its disclosure rules until after the crisis, despite substantial changes in macroeconomic conditions in the prior decade. We interpret this as a commitment to the disclosure policy instituted during the economic modernization that followed the 1982 crisis. The durability of the post-1995 weekly reporting regime reinforces that interpretation. Beyond commitment to the disclosure policy, we follow [Horn et al. \(2022\)](#) and evaluate the

precision of reserve reports, finding them to be accurate. In short, disclosure was infrequent, scheduled, and reliable for over a decade; only after the 1994-95 crisis did Mexico undertake a one-time, durable revision of its disclosure policy.

Related Literature. Our paper is related to three strands of literature. The first of these is the theoretical literature on sovereign debt initiated by [Eaton and Gersovitz \(1981\)](#), and [Cole and Kehoe \(2000\)](#), among others. Unlike the aforementioned papers, our model accounts for incomplete information as in [Sandleris \(2008\)](#), [Perez \(2017\)](#), [Amador and Phe-
lan \(2021\)](#), and others. [Zabai \(2014\)](#) and [Szkup \(2022\)](#) also work in incomplete information environments, but differently from the previous papers, they rely on dispersed information to get rid of the equilibrium multiplicity commonly observed under the Cole-Kehoe timing assumption. [Bassetto and Galli \(2019\)](#) study sovereign default in an environment in which the currency choice of debt affects debt prices as it shifts information held by marginal investors. We differ from this literature by introducing an endogenous choice of disclosure in the sovereign's problem, making lenders' information (or the lack thereof) an endogenous object in our model.

We are related to the sovereign-debt literature on *hidden debt* and transparency, including [Gu and Stangebye \(2023\)](#), [Gamboa \(2023\)](#), [Guler et al. \(2022\)](#), and [Horn et al. \(2022\)](#). We differ by focusing on a distinct source of information asymmetry: limited transparency about the Mexican central bank's U.S. dollar reserves (rather than undisclosed liabilities). Among the closest antecedents, [Guler et al. \(2022\)](#) quantifies the macro effects of disclosing non-Paris Club debt in a quantitative model with *exogenous* information regimes, finding modest welfare losses from transparency; [Horn et al. \(2022\)](#) studies hidden debt and transparency with *lender monitoring* that endogenously triggers revelations and affects prices and default. Our framework departs in two key ways. First, the object of opacity is reserves, which is conceptually parallel in our model because hidden debt due imminently maps into our type (repayment capacity). Second, and more importantly, we assume the sovereign can *commit ex ante to a disclosure policy*, giving it a credible channel to communicate with lenders; this contrasts with [Horn et al. \(2022\)](#) environment where lenders choose whether to monitor. This shift in the institutional locus of transparency makes the information regime itself a policy choice.

Second, we contribute to the debate on how information disclosure affects crises. [Gorton and Ordoñez \(2020\)](#) studies how to reveal information *during* an ongoing crisis to manage it in equilibrium. In contrast, in our model, the government chooses the information rule *ex ante*, before a crisis materializes - crucially, in our framework, the disclosure choice itself

can *generate or avert* a crisis. This places us with the pre-crisis design literature that treats disclosure as a commitment device: stress test design in [Goldstein and Leitner \(2018\)](#) and disclosure to prevent runs in [Faria-e Castro et al. \(2017\)](#).¹ Despite the banking-sovereign differences - and our departure from multiplicity toward a unique pooling outcome under non-disclosure - the insurance logic is closely related to [Goldstein and Leitner \(2018\)](#).

Finally, our paper is related to the information design literature: [Kamenica and Gentzkow \(2011\)](#), [Bergemann et al. \(2015\)](#), [Inostroza and Pavan \(2023\)](#).² More specifically, we focus on a choice of disclosure in environments with adverse selection, as in [Garcia and Tsur \(2021\)](#), [Immorlica et al. \(2022\)](#), and [Dovis and Martellini \(2024\)](#). Differently from them, we focus on the role of disclosure policies in a debt crisis episode, the Mexican Peso Crisis.

Outline. The rest of the paper is organized as follows. In Section (2) we present our environment and the results. Section (3) provides a background on the Mexican Peso Crisis (1994-1995), and relates our model to the behavior of the Mexican government before and during the crisis. Section (4) concludes.

2 Model

Environment. Time is discrete and indexed by $t \in \{0, 1, 2\}$. The economy is populated by two types of agents: a risk-averse sovereign and a continuum of risk-neutral competitive investors. The sovereign derives utility from consuming the only good in the economy at $t = 1, 2$ and discounts the future at a rate $\beta \in (0, 1)$. R is the constant and exogenous risk-free rate. At $t = 0$ the expected payoff of the sovereign is given by:

$$\mathbb{E} [u(c_1) + \beta u(c_2)]$$

Throughout this section, we will assume $u(x) = \log(x)$. The expectation is taken with respect to the sovereign's period-2 endowment θ .

$$\theta = \begin{cases} \theta_L, & \text{with prob. } \pi \\ \theta_H, & \text{with prob. } 1 - \pi \end{cases}$$

At $t = 1$ the sovereign learns its type (θ), consumes using a constant endowment y , and new debt (b) is issued at a price q . For simplicity, we assume that the sovereign has no

¹For a survey, see [Goldstein and Yang \(2017\)](#).

²For a detailed survey, see [Bergemann and Morris \(2019\)](#).

debt due in the first period. Under these assumptions,

$$c_1 = y + qb$$

The sovereign cannot commit ex-ante to repay at $t = 2$. In case of repayment, the endowment θ is used to repay the debt and consume. In case of default, the sovereign uses its default endowment y_d to consume.

$$c_2 = \delta \times y_d + (1 - \delta) \times (\theta - b)$$

Where $\delta \in \{0, 1\}$ is equal to 1 if the sovereign defaults, and zero otherwise.

Information and disclosure. At $t = 0$ the sovereign commits to a disclosure policy ϕ . At $t = 1$ the sovereign is privately informed about its next period endowment $\theta \in \{\theta_H, \theta_L\}$ and the disclosure policy is realized. Lenders have a well-specified prior, that is, they know the true probability distribution over θ , and update their beliefs about the sovereign's type after observing the sovereign's choice of debt. We restrict our analysis to these extremes: fully revealing θ (full-disclosure) or not revealing it at all (no-disclosure), that is $\phi \in \{0, 1\}$. Figure (2) exhibits a diagram with the timing.

Figure 2: Timing of events

$t = 0$	$t = 1$	$t = 2$
1) Choice of disclosure policy $\phi \in \{0, 1\}$	1) Government learns its type (θ_i) 2) Govt type is revealed ($\phi = 1$) or not ($\phi = 0$) 3) Choice of debt 4) Price is determined q 5) Consumption takes place	1) Default decision 2) Consumption

Discussion. We analyze Mexico's reserve disclosure policy within our framework, mapping 1982-1995 to the first play of our model and 1995-present to the second.

For the first play, we take 1982 as $t = 0$. At $t = 0$, the government reaffirmed an infrequent, scheduled reserve-disclosure rule and kept it unchanged until the end of the first play. At $t = 1$ (1983-1993), Mexico borrowed dollars under imperfect information: lenders observed announcements but not true reserves, so pricing reflected beliefs about

repayment capacity rather than full information. The crisis year 1994 is $t = 2$, when the default decision is realized. We interpret y as the U.S. dollars available to the government net of maturing obligations and the outlays required to defend the peg; θ denotes reserves after peg-related outflows available to repay debt at $t = 2$.

In 1995, the game restarts for its second play: the country chooses its disclosure policy again under higher default costs - an assumption we justify by examining changes in IMF conditionality and program design - and adopts the weekly reporting regime maintained since.

2.1 Complete Information Benchmark

We start by stating the game's equilibrium under complete information, we then characterize the equilibrium and compute the sovereign's expected payoff under full-disclosure ($\phi = 1$). Under complete information, the payoff of a type- θ , given the default endowment (y_d) sovereign is given by:

$$v_{FD}(\theta) = \max_{b(\theta)} \left\{ \log(y + q(\theta, b(\theta))b(\theta)) + \beta \log(c_2^*(b, \theta)) \right\} \quad (1)$$

subject to:

$$c_2^*(b, \theta) = \max_{\delta \in \{0,1\}} \delta \times y_d + (1 - \delta) \times (\theta - b)$$

Definition 1. An *equilibrium under complete information* is a strategy for the sovereign ($\delta^*(b, \theta), b_{FD}^*(\theta)$), a price schedule $q_{FD}(b, \theta)$ such that:

- i) The default policy $\delta^*(b, \theta)$ maximizes consumption at $t = 2$
- ii) Given $q_{FD}(b, \theta)$, the choice of debt $b_{FD}^*(\theta)$ maximizes the sovereign's lifetime payoff
- iii) Investors make zero profits

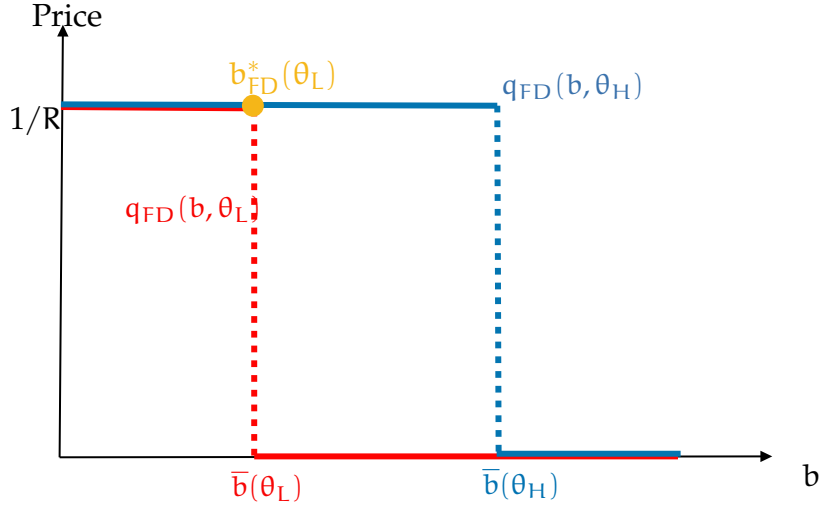
$$\frac{1}{R} \left[\delta(b, \theta) \times 0 + (1 - \delta(b, \theta)) \right] - q(b, \theta) = 0$$

Note that under complete information, the strategies and prices are explicit functions of the sovereign's type. In particular, Figure (3) shows the two price schedules faced by each type of sovereign. We denote the choice of debt of a type- θ sovereign as $b_{FD}^*(\theta)$. Before proceeding with the characterization of equilibrium we state some parametric assumptions:

Assumption 1 (Constrained L-type).

$$y_d > \frac{\beta R}{1 + \beta} \left(\frac{\theta_L}{R} + y \right)$$

Figure 3: Debt and prices in the FD equilibrium (with Assumption 1)



Assumption (1) ensures that the L-type sovereign chooses as much debt as lenders are willing to lend at the risk-free price $1/R$, as shown in Figure (3). This makes L-type sovereigns indifferent between defaulting and repaying at $t = 2$. When this is the case we say that the L-type is constrained.

Lemma 1. Define $\bar{b}(\theta) \equiv \theta - y_d$. The following strategy is an *equilibrium*:

$$\delta^*(b, \theta) = \begin{cases} 1, & \text{if } b > \bar{b}(\theta) \\ 0, & \text{if } b \leq \bar{b}(\theta) \end{cases}$$

$$q_{FD}(b, \theta) = \begin{cases} 0, & \text{if } b > \bar{b}(\theta) \\ 1/R, & \text{if } b \leq \bar{b}(\theta) \end{cases}$$

$$b_{FD}^*(\theta) = \min \left\{ \frac{\theta - \beta R y}{1 + \beta}, \bar{b}(\theta) \right\}$$

Proof. See Appendix (A.1) □

Finally, we define the expected payoff of a sovereign under complete information induced

by a full-disclosure policy, given y_d as:

$$\begin{aligned}
V_{FD}(\pi) \equiv & \pi \times \underbrace{\left[u(y + q_{FD}(b_{FD}^*(\theta_L), \theta_L) b_{FD}^*(\theta_L)) + \beta u(\theta_L - b_{FD}^*(\theta_L)) \right]}_{v_{FD}(\theta_L; y_d)} \\
& + (1 - \pi) \times \underbrace{\left[u(y + q_{FD}(b_{FD}^*(\theta_H), \theta_H) b_{FD}^*(\theta_H)) + \beta u(\theta_H - b_{FD}^*(\theta_H)) \right]}_{v_{FD}(\theta_H; y_d)} \\
& \hspace{15em} \text{(Exp. Payoff of FD)}
\end{aligned}$$

By choosing and committing to fully disclose its type, the sovereign gets a lottery-like payoff in which price schedules, debt, and default policies as described in Lemma (1) are weighted by the probability of each of the types to be realized.

2.2 Incomplete Information

We now consider a version of the game with incomplete information. In this game the sovereign chooses not to disclose its type to the lenders ($\phi = 1$). Lenders now observe the choice of debt and update their prior π to compute *posterior beliefs* of the sovereign's type. We denote the posterior belief conditioned on an observed debt choice as $\mu(b) \equiv \mathbb{P}(\theta = \theta_L | b)$. The sovereign's problem is given by:

$$v_{ND}(\theta; y_d) = \max_{b(\theta)} \left\{ \log(y + q(b(\theta))b(\theta)) + \beta \log(c_2^*(b, \theta)) \right\} \quad (2)$$

subject to:

$$c_2^*(b, \theta) = \max_{\delta \in \{0,1\}} \delta \times y_d + (1 - \delta) \times (\theta - b)$$

Definition 2. A *Perfect Bayesian Equilibrium (PBE)* is a strategy for the sovereign $(b_{ND}^*(\theta), \delta^*(b, \theta))$, a price schedule $q_{ND}(b)$, and beliefs $\mu(b)$ such that:

1. The default policy $\delta^*(b, \theta)$ maximizes the sovereign's consumption at $t = 2$.
2. Given $q_{ND}(b)$ the choice of debt maximizes sovereign's lifetime payoffs.
3. Beliefs follow the Bayesian update whenever possible
4. Investors make zero profits

$$\frac{1}{R} \mathbb{E}_\mu \left[\delta(b, \theta) \times 0 + (1 - \delta(b, \theta)) \right] - q(b, \theta) = 0$$

We will look for equilibria in pure strategies³. Note that at $t = 2$ as before, the default choice of the sovereign depends only on the amount of debt chosen at $t = 1$ and on the sovereign's type. We will show that the default threshold strategies will be the same as in the full-disclosure equilibrium. This follows from the fact that all action generated by the lack of information in our model happens at $t = 1$, through prices and debt choices. The main conceptual difference in this case follows from the fact that the lender's strategy is not conditioned on types, since those are not observed.

Definition 3. A *separating equilibrium* is a PBE in which both types pick different levels of debt: $b_{ND}^*(\theta_L) \neq b_{ND}^*(\theta_H)$

Proposition 1. Under *Assumption (1)*, there is no separating equilibrium.

Proof. See Appendix (A.2) □

This result resembles the classic result of Nachman and Noe (1994). In our framework Assumption (1) guarantees that the optimal debt chosen by the L-type lies in a region where they would default, while the H-type repays, the “*risky region*”⁴. Under this condition, we rule out the existence of separating equilibria. If Assumption (1) is dropped, a separating equilibrium may emerge in which the L-type borrows its optimal amount at price $1/R$, while the H-type separates by choosing a debt level that is costly for itself.

Definition 4. A *pooling equilibrium* is a PBE in which both types pick the same debt level: $b_{ND}^*(\theta_L) = b_{ND}^*(\theta_H) = \tilde{b}$

We now focus on two natural candidates for the common debt level $\tilde{b}$ chosen in a pooling equilibrium. We refer to these as *risky pooling* and *safe pooling*. In a risky pooling equilibrium, both types choose $\tilde{b} > \theta_L - y_d$, but the L-type defaults at $t = 2$ while the H-type repays in full. In a safe pooling equilibrium, both types choose the lower debt level $\theta_L - y_d$, which is repaid in full by both types. The distinction is relevant because the risky pooling outcome sustains a higher common debt level, at the cost of default by the L-type, while the safe pooling outcome avoids default but entails lower borrowing. These two cases will guide the characterization of pooling equilibria. Define the following debt

³The sovereign will pick one debt level, instead of a probability distribution of possible debt levels.

⁴This is equivalent to the assumption that firms cannot finance themselves entirely from riskless debt. Nachman and Noe (1994) state their result for an environment with multiple types and impose the strict stochastic dominance (SSD) condition; in our two-type case, this is reduced to $\theta_L < \theta_H$. In the two-type case, Nachman and Noe (1994) shows that all equilibria are pooling.

level:

$$\tilde{b}(\pi) \equiv \max \left\{ \theta_L - y_d, \min \left\{ \frac{\theta_H - \frac{\beta R y}{1-\pi}}{1+\beta}, \theta_H - y_d \right\} \right\}$$

Let the belief system, the default policy, and the price schedule be:

$$\mu(b; \pi) = \begin{cases} \pi, & \text{if } b \leq \tilde{b}(\pi) \\ 1, & \text{if } b > \tilde{b}(\pi) \end{cases}$$

$$\delta^*(b, \theta) = \begin{cases} 1, & \text{if } b > \bar{b}(\theta) \\ 0, & \text{if } b \leq \bar{b}(\theta) \end{cases}$$

$$q_{ND}(b) = \begin{cases} 0, & \text{if } b > \theta_H - y_d \\ \frac{1-\mu(b)}{R}, & \text{if } b \in (\theta_L - y_d, \theta_H - y_d] \\ 1/R, & \text{if } b \leq \theta_L - y_d \end{cases}$$

We justify our choice of on- and off-path beliefs via a simple mechanism design problem in Appendix (A.3) that identifies the H-type preferred pooling PBEs and delivers exactly the two candidates we study: *risky pooling* (the L-type defaults, the H-type repays) and *safe pooling* (both types repay). Define the interval for the default endowment that makes the L-type the only constrained type when facing a price $1/R$:

$$Y_d \equiv \left(\frac{R\beta}{1+\beta} \left(y + \frac{\theta_L}{R} \right), \frac{R\beta}{1+\beta} \left(y + \frac{\theta_H}{R} \right) \right)$$

Proposition 2. *Let $y_d \in Y_d$. There exists $\pi^*(y_d)$ such that for $\pi < \pi^*(y_d)$, the unique H-type preferred PBE is the risky pooling equilibrium. Moreover,*

$$\frac{\partial \pi^*(y_d)}{\partial y_d} > 0$$

Proof. See Appendix (A.4) □

In a pooling outcome, lenders discount the price to reflect the chance of facing an L-type. Because the H-type repays, it bears this discount and therefore benefits from a higher

price. When the prior probability of the L-type, π , is small, the lenders belief of meeting an L-type is small, the price discount is mild (close to the risk-free price), and the H-type can place more debt on favorable terms. Hence, for sufficiently small π (below some cutoff π^*), the H-type strictly prefers risky pooling to the safe cap $\theta_L - y_d$. Intuitively, to induce the H-type to pool with a defaulter, beliefs must put little weight on the L-type, precisely what happens when π is low. Also, since the H-type's safe pooling payoff is decreasing on the default endowment, low levels of y_d would make the H-type ask for higher pooling prices (lower π) to sustain a risky pooling.

We can then compute the expected payoff under a non-disclosure policy as:

$$V_{ND}(\pi) \equiv u\left(y + \frac{1-\pi}{R} \times \tilde{b}(\pi)\right) + \beta[\pi u(y_d) + (1-\pi)u(\theta_H - \tilde{b}(\pi))]$$

Notice that the above equation represents the payoff under no-disclosure for $\pi < \pi^*(y_d)$, when the *risky pooling* equilibrium is played, and $y_d \in Y_d$, that is, only the L-type is constrained when facing a price $1/R$ ⁵. Define the difference between the no-disclosure (ND) and full disclosure (FD) payoffs, such as:

$$\Delta(\pi) \equiv V_{ND}(\pi) - V_{FD}(\pi)$$

Before stating our main result, we will state a condition on the parameters.

$$\log\left(\frac{Ry + \theta_H}{Ry + \theta_L}\right) + \frac{R\beta y - \theta_H}{Ry + \theta_H} > 0 \quad (\star)$$

Notice that the first term in the above expression shows the ratio between the $t = 2$ endowment of the two types. This condition is satisfied for large gaps between the endowments of the two types ($\theta_H - \theta_L$), that is, if the H-type is wealthier enough than the L-type. If that is the case, we can state our main result: for any default endowment, there exists an interval of beliefs in which the government chooses not to disclose information to the lenders.

Proposition 3. Suppose $(\star)$, holds. Then, for any $y_d \in Y_d$, there exists $\epsilon(y_d) > 0$ such that

$$\Delta(\pi) > 0, \quad \text{for all } \pi \in (0, \epsilon(y_d))$$

⁵We are not interested in the parametrizations in which the *safe pooling* equilibrium is played. In this equilibrium, the H-type is able to costly signal its repayment capacity by choosing a debt level dominated by its own debt level under full disclosure. The case where the H-type is constrained is simple and treated in the Appendix (A.6).

That is, non-disclosure (ND) is preferred over full disclosure (FD).

Proof. We start by rewriting $\Delta(\pi)$:

$$\begin{aligned}\Delta(\pi) = & \log \left(\frac{y + \frac{1-\pi}{R}\theta_H}{1+\beta} \right) + \beta(1-\pi) \log \left(\frac{R\beta}{1+\beta} \left(\frac{y}{1-\pi} + \frac{\theta_H}{R} \right) \right) - \pi \log \left(y + \frac{\theta_L - y_d}{R} \right) \\ & - (1-\pi) \left[\log \left(\frac{y + \frac{\theta_H}{R}}{1+\beta} \right) + \beta \log \left(\frac{R\beta}{1+\beta} \left(y + \frac{\theta_H}{R} \right) \right) \right]\end{aligned}$$

Notice that $\Delta(0) = 0$. Differentiating $\Delta(\pi)$ with respect to π , and evaluating it at zero gives us the following expression:

$$\Delta'(0) = \log \left(\frac{Ry + \theta_H}{(1+\beta)(Ry + \theta_L - Ry_d)} \right) + \frac{R\beta y - \theta_H}{Ry + \theta_H}$$

Notice that $\Delta'(0)$ depends on y_d only through $-\log(Ry + \theta_L - Ry_d)$, which makes $\Delta'(0, y_d)$ increasing in y_d . This means that if $\Delta'(0) > 0$ for the minimum admissible value of $y_d \in Y_d$ it must be greater than zero for all admissible values of y_d . Recall that the minimum admissible value of y_d , denoted by $y_d^{\min}$ is:

$$y_d^{\min} = \frac{R\beta}{1+\beta} \left(y + \frac{\theta_L}{R} \right)$$

We then get condition ($\star$):

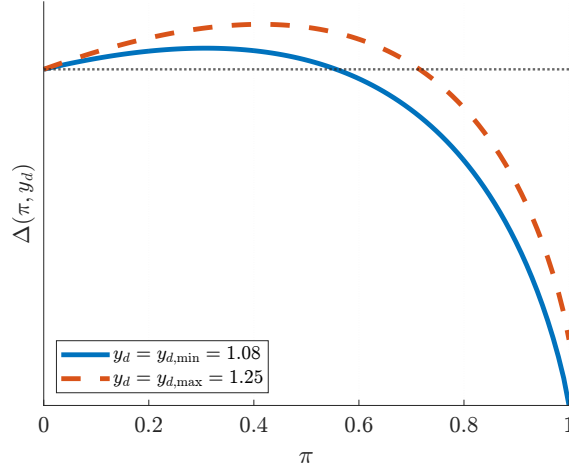
$$\Delta'(0; y_d^{\min}) = \log \left(\frac{Ry + \theta_H}{Ry + \theta_L} \right) + \frac{R\beta y - \theta_H}{Ry + \theta_H} > 0$$

□

Notice that we must restrict the above result to values of $\pi < \pi^*(y_d)$. However, it must be the case around $\pi = 0$, which is the point used in our proof. This also explains why non-disclosure is appealing for small π : lenders belief of facing an L-type is tiny, so the pooling price is only mildly discounted (close to $1/R$). The H-type then bears little pooling cost and can borrow on near risk-free terms, while the L-types second-period outcome is unchanged by default and enjoys higher first-period borrowing under pooling. This would make the decision of not disclosing more attractive *ex-ante*. Condition ($\star$) ensures that the L-type's marginal insurance gains exceed the H-type's pooling cost at $\pi = 0$.

As π rises, the price discount deepens, the H-types pooling cost grows, and disclosure

Figure 4: Difference between expected payoffs under ND and FD, for different beliefs.



Note: Parameters used: $\beta = 0.92$, $R = 1$, $y = 1$, $\theta_L = 1.25$, $\theta_H = 5$.

becomes relatively more attractive. Figure (4) exhibits regions in which ND is preferred ($\Delta(\pi) > 0$) for the minimum and maximum admissible values for $y_d \in Y_d$.

In Appendix (A.5), we elaborate on how our commitment assumption relates to the monitoring problem in Horn et al. (2022). We also perform an additional comparative statics, showing that under commitment, the sovereigns incentives to fully disclose its state shrink in bad times, consistent with the idea that lenders choose to monitor the government more closely during such periods, triggering hidden debt revelations.

3 Mexican Peso Crisis 1994

In this section, we justify the main assumptions of our model and explain how the Mexican crisis of 1994 can be analyzed through the lens of the model.

3.1 Historical Background: Commitment Through Reform and Turmoil

After the 1982 crisis, Mexico underwent a profound economic modernization. The government implemented fiscal consolidation, reducing public expenditure from 45% of GDP in 1983 to 27% in 1993, and privatized more than 75% of state-owned enterprises. Trade liberalization began in 1983; by 1985, tariffs on over 85% of imports had been eliminated. These reforms attracted large foreign capital inflows in the early 1990s and kept the peso strong against the U.S. dollar. Until 1994, Banco de México successfully accumulated international reserves, and maintaining an appreciated peso was not seen as a major chal-

lenge.

In 1994, however, a sequence of political and social shocks undermined investor confidence. For instance, the *Ejercito Zapatista de Liberacion Nacional* (EZLN) uprising, high-profile political violence including the assassination of presidential candidate Luis Donaldo Colosio, the resignation of the interior minister, and the later assassination of José Francisco Ruiz Massieu. Despite this turbulent backdrop and the substantial macroeconomic changes of the prior decade, the government did not revisit its reserve disclosure policy. Pre-crisis disclosure remained infrequent but scheduled and reliable, and the rule was only revised after the 1994-95 crisis.

3.2 Truth-telling

We assume the sovereign can commit ex ante to a disclosure rule about a fundamental. When the fundamental is directly policy-controlled, such as public debt or GDP the sovereign may have incentives to misreport or later revise disclosures to lower borrowing costs, ease rollover risk, or avoid scrutiny; hidden debt is a salient example (Horn et al., 2022). We ask whether this logic applies to reserves by examining the institution responsible for documenting them before the crisis (Banco de México) and its reporting practices.

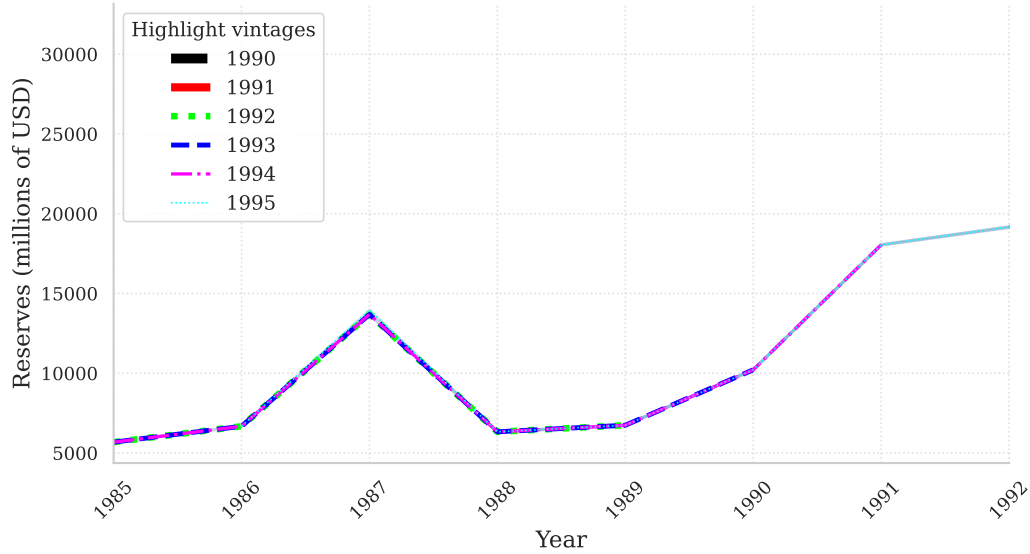
We use multiple vintages from the World Banks *Global Development Finance* and *The World Debt Tables*⁶ to assess reserve reports. Across vintages, reserve levels were reported accurately before, during, and after the crisis. Because IDS relies on direct debtor reporting, omissions and revisions can be traced to sovereign reporting decisions; an upward revision indicates that a previously unreported item is now recorded.⁷

Consistent with truthful reserve reporting, tracing the *same calendar year* across different vintages yields essentially identical reserve values: in Figure (5), the highlighted vintage lines lie on top of one another. This overlap indicates negligible across-vintage revisions of reserves and supports the assumption that, whenever reserves are disclosed, the reported value is the truth. Such findings are consistent with what we show in the right panel of Figure (1), press-reported values about past values of reserves lie mostly on the 45-degree line.

⁶Before 2013, the *International Debt Statistics* (IDS) was named *Global Development Finance*. Publications appeared as Volume I (analysis and commentary) and Volume II (summary and country tables). Before 1997, analysis and tables were published as *The World Debt Tables*.

⁷Ex post revisions do not reflect market-value fluctuations; in the debt series, instruments are recorded at face value.

Figure 5: Reports about the same calendar years, across vintages



3.3 IMF conditionality and the costs of default

Our model predicts that when the cost of default is higher the region in which non-disclosure is optimal shrinks. We proxy default costs with IMF program stringency - measured by the number of conditions per arrangement- as in [Kentikelenis et al. \(2016\)](#) and [Akerman et al. \(2022\)](#). Figure (6) shows a clear increase in the average number of conditions after 1995.

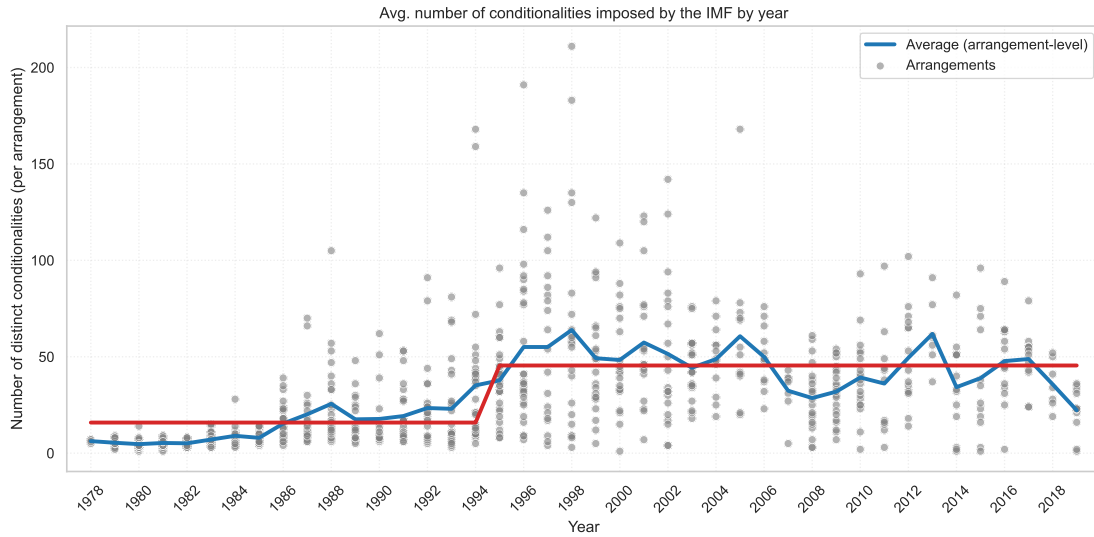


Figure 6: Conditions imposed by the IMF in deals pre/post 1995.

To formalize the pattern in the figure above we estimate:

$$\text{Number of conditions}_t = \beta_0 + \beta_1 \text{Post-1995}_t + \epsilon_t,$$

where the dependent variable is the average count of conditions on IMF arrangements approved in year t , and Post-1995_t is a binary variable indicating whether t is before or after 1995.⁸ The results indicate that, on average, arrangements after 1995 feature about 45 conditions versus roughly 15 before 1995.

Mexico maintained the same reserve-disclosure rule through the 1980s and did not revisit it until the 1994-95 crisis, after which it adopted weekly reserve reporting. Through the lens of our model, the post-1995 increase in default costs makes full disclosure optimal at the start of the second play of the game, consistent with the observed shift to weekly reporting.

4 Conclusion

We develop an analytic sovereign-default model in the Eaton-Gersovitz tradition with incomplete information, where the government chooses, *ex ante*, whether to disclose its repayment capacity to lenders. Under no disclosure, there are no separating equilibria; instead, a *risky pooling* equilibrium emerges in which both types issue the same face value, the L-type receives a positive price yet defaults in period 2, and the H-type repays but bears a price discount due to adverse selection. Commitment matters: by committing not to disclose, the sovereign secures insurance across states via this cross-subsidization. We show that when deadweight losses from default are small, the set of lender priors for which no disclosure is preferred is larger (equivalently, as default becomes more costly, the region favoring no disclosure shrinks and disclosure becomes relatively more attractive).

We read Mexico's 1994-95 episode through this lens. Prior to the crisis, updated information on dollar reserves was not made available at high frequency, yet vintage comparisons show reserve reports were essentially exact. This supports the assumption that, conditional on disclosure, reserves were truthfully reported under a long-standing, rule-like regime before the crisis, followed by a durable shift to weekly reporting afterward. To gauge changes in default costs, we use IMF conditionality: the number of conditions attached to Mexico's arrangements rises after 1995, and a simple post-1995 regression con-

⁸Dataset: <https://www.imfmonitor.org>.

firms a significant step-up. Interpreted in our framework, higher post-crisis default costs rationalize the move toward fuller, higher-frequency disclosure at the start of the second play of the game.

References

- Akerman, Ariel, João Paulo Pessoa, and Leonardo Weller**, “The Wests teeth: IMF conditionality during the Cold War,” *The World Economy*, 2022, 45 (7), 2034–2051.
- Amador, Manuel and Christopher Phelan**, “Reputation and sovereign default,” *Econometrica*, 2021, 89 (4), 1979–2010.
- Bassetto, Marco and Carlo Galli**, “Is inflation default? The role of information in debt crises,” *American Economic Review*, 2019, 109 (10), 3556–3584.
- Bergemann, Dirk and Stephen Morris**, “Information design: A unified perspective,” *Journal of Economic Literature*, 2019, 57 (1), 44–95.
- , **Benjamin Brooks, and Stephen Morris**, “The limits of price discrimination,” *American Economic Review*, 2015, 105 (3), 921–957.
- Cole, Harold L and Timothy J Kehoe**, “Self-fulfilling debt crises,” *The Review of Economic Studies*, 2000, 67 (1), 91–116.
- Crawford, Vincent P and Joel Sobel**, “Strategic information transmission,” *Econometrica: Journal of the Econometric Society*, 1982, pp. 1431–1451.
- Dovis, Alessandro and Paolo Martellini**, “Long-Term Contracts, Commitment, and Optimal Information Disclosure,” 2024.
- e Castro, Miguel Faria, Joseba Martinez, and Thomas Philippon**, “Runs versus lemons: information disclosure and fiscal capacity,” *The Review of Economic Studies*, 2017, 84 (4), 1683–1707.
- Eaton, Jonathan and Mark Gersovitz**, “Debt with potential repudiation: Theoretical and empirical analysis,” *The Review of Economic Studies*, 1981, 48 (2), 289–309.
- Gamboa, Juliana**, “Hidden chinese lending,” *Available at SSRN* 4617798, 2023.
- Garcia, Daniel and Matan Tsur**, “Information design in competitive insurance markets,” *Journal of Economic Theory*, 2021, 191, 105160.
- Goldstein, Itay and Liyan Yang**, “Information disclosure in financial markets,” *Annual Review of Financial Economics*, 2017, 9, 101–125.
- **and Yaron Leitner**, “Stress tests and information disclosure,” *Journal of Economic Theory*, 2018, 177, 34–69.

- Gorton, Gary and Guillermo Ordoñez**, "Fighting crises with secrecy," *American Economic Journal: Macroeconomics*, 2020, 12 (4), 218–245.
- Gu, Grace Weishi and Zachary R Stangebye**, "Costly information and sovereign risk," *International Economic Review*, 2023, 64 (4), 1397–1429.
- Guler, Bulent, Yasin Kürşat Önder, and Temel Taskin**, "Asymmetric Information and Sovereign Debt Disclosure," *Available at SSRN 4048018*, 2022.
- Horn, Sebastian, David Mihalyi, Philipp Nickol, and César Sosa-Padilla**, "Hidden debt revelations," *Unpublished. World Bank, Washington, DC*, 2022.
- Immorlica, Nicole, Andre M Sztutman, and Robert M Townsend**, "Optimal Credit Scores Under Adverse Selection," in "Proceedings of the 23rd ACM Conference on Economics and Computation" 2022, pp. 737–738.
- Inostroza, Nicolas and Alessandro Pavan**, "Adversarial coordination and public information design," *Available at SSRN 4531654*, 2023.
- Kamenica, Emir and Matthew Gentzkow**, "Bayesian persuasion," *American Economic Review*, 2011, 101 (6), 2590–2615.
- Kentikelenis, Alexander E, Thomas H Stubbs, and Lawrence P King**, "IMF conditionality and development policy space, 1985–2014," *Review of International Political Economy*, 2016, 23 (4), 543–582.
- Nachman, David C and Thomas H Noe**, "Optimal design of securities under asymmetric information," *The Review of Financial Studies*, 1994, 7 (1), 1–44.
- Perez, Diego J**, "Sovereign debt maturity structure under asymmetric information," *Journal of International Economics*, 2017, 108, 243–259.
- Sandleris, Guido**, "Sovereign defaults: Information, investment and credit," *Journal of international Economics*, 2008, 76 (2), 267–275.
- Szkup, Michal**, "Preventing self-fulfilling debt crises: A global games approach," *Review of Economic Dynamics*, 2022, 43, 22–55.
- Zabai, Anna**, "Managing default risk," 2014.

A Appendix

A.1 Proof of Lemma 1

Default decision. At $t = 2$ the sovereign decides whether to default or repay on its debt b to maximize consumption at $t = 2$. We denote the default decision by $\delta^*(\theta, b) \in \{0, 1\}$:

$$\delta^*(b, \theta) = \arg \max_{\delta \in \{0, 1\}} \delta \times y_d + (1 - \delta) \times (\theta - b) \quad (3)$$

The default policy at $t = 2$ depends solely on the amount of debt picked at $t = 1$, and the sovereign's type:

$$\delta^*(b, \theta) = \begin{cases} 1, & \text{if } b > \bar{b}(\theta) \\ 0, & \text{if } b \leq \bar{b}(\theta) \end{cases}$$

$$y_d = \theta - \bar{b}(\theta)$$

Where $\bar{b}(\theta) \equiv \theta - y_d$ are the upper thresholds for debt levels that make each type indifferent between repaying and defaulting. The default policy determines consumption at $t = 2$:

$$c_2^*(b, \theta) = \delta^*(b, \theta) \times y_d + [1 - \delta^*(b, \theta)] \times (\theta - b)$$

Lenders' problem. The strategy for the lenders is a price schedule $q_{ND}(b, \theta)$:

$$q_{FD}(b, \theta) = \begin{cases} 0, & \text{if } b > \bar{b}(\theta) \\ 1/R, & \text{if } b \leq \bar{b}(\theta) \end{cases}$$

The following expression gives the interior solution for debt:

$$\frac{\theta - \beta R y}{1 + \beta}$$

However, conditional on repayment the sovereign's choice of debt can be a corner solution:

$$b(\theta) = \min \left\{ \frac{\theta - \beta R y}{1 + \beta}, \theta - y_d \right\}$$

□

A.2 Proof of Proposition 1

Let $b_{ND}^*(\theta_H)$ and $b_{ND}^*(\theta_L)$ be the separating choices of debt. By definition, in a separating equilibrium, lenders can distinguish between types, that is, $\mu(b_{ND}^*(\theta_L)) = 1$ and $\mu(b_{ND}^*(\theta_H)) = 0$. By the zero profit condition, it must be the case that debt is priced at $q_{ND}(b) \in \{0, 1/R\}$ for $b \in \{b_{ND}^*(\theta_L), b_{ND}^*(\theta_H)\}$.

Claim 1. *For any type θ , we must have that $q_{ND}(b_{ND}^*(\theta)) > 0$*

Proof. Suppose not, that is, there exists $\theta^* \in \{\theta_H, \theta_L\}$ such that $q_{ND}(b_{ND}^*(\theta^*)) = 0$. It then must be the case that $b_{ND}^*(\theta^*) > \theta^* - y_d$ which gives the utility:

$$\begin{aligned} u(y) + \beta u(y_d) &< \underbrace{u\left(y + \frac{\theta_L - y_d}{R}\right) + \beta u(y_d)}_{\text{L-type's payoff by picking } b_{ND}^*(\theta_L) = \theta_L - y_d} \\ u(y) + \beta u(y_d) &< \underbrace{u\left(y + \frac{\theta_L - y_d}{R}\right) + \beta u(\theta_H - \theta_L + y_d)}_{\text{H-type's payoff by picking } b_{ND}^*(\theta_H) = \theta_L - y_d} \end{aligned}$$

We thus conclude that any choice of debt that induces zero price (and default) cannot be our separating equilibrium. $\square$

By **Assumption (1)** the L-type sovereign $\theta_L - y_d$ is preferred over any $b < \theta_L - y_d$. It then must be the case that $b_{ND}^*(\theta_L) = \theta_L - y_d$, and the L-type's utility at $t = 2$ is given by $u(y_d)$. If the L-type is constrained at $\theta_L - y_d$, it must be the case that the H-type prefers $\theta_L - y_d$ to any debt level smaller than this.

Our candidate for separating equilibrium must be such that $b_{ND}^*(\theta_H) \in (\theta_L - y_d, \theta_H - y_d)$, that is, the H-type's choice of debt is in the default region for the L-type, a region where the L-type's payoff is strictly increasing on its debt level. Moreover by the above claim: $q_{ND}(b_{ND}^*(\theta_H)) = 1/R$. Therefore:

$$u\left(y + \frac{\theta_L - y_d}{R}\right) + \beta u(y_d) < u\left(y + \frac{b_{ND}^*(\theta_H)}{R}\right) + \beta u(y_d)$$

We thus conclude that $b_{ND}^*(\theta_H)$ is a profitable deviation for the L-type regardless of the belief system. $\square$

A.3 Mech. Design Approach: H-type preferred pooling PBE

This section considers a mechanism design problem in which the designer is the H-type sovereign. We start by defining some useful objects:

$$v_L^a = u(y) + \beta u(\theta_L) \quad (4)$$

We split the problem in two: (i) investors break even for each of the types; (ii) investors pick a pooling contract in which the L-type defaults.

A.3.1 The Best Outcome with Default

We now consider a problem in which the designer offers a pooling contract and the L-type defaults. In this case, ICR-L (Incentive Compatibility Repayment for the L-type) is not required further and ICD-L (Incentive Compatibility of Default for L-type) is reduced to $\tilde{b} > \theta_L - y_d$. Since investors are offering a pooling contract, its participation constraint (PC-I) takes into account the probability of facing each of the types.

$$\max_{\{\tilde{b}, \tilde{q}\}} u(y + \tilde{q}\tilde{b}) + \beta u(\theta_H - \tilde{b}) \quad (P)$$

subject to

$$\tilde{b} \leq \theta_H - y_d \quad (\text{SUST-H}, \nu)$$

$$\tilde{b} \geq \theta_L - y_d \quad (\text{ICD-L}, \chi)$$

$$u(y + \tilde{q}\tilde{b}) + \beta u(y_d) \geq v_L^a \quad (\text{PC-L}, \eta)$$

$$\underbrace{\pi \times [-\tilde{q}\tilde{b}] + (1 - \pi) \times \left[\frac{\tilde{b}}{R} - \tilde{q}\tilde{b} \right]}_{\frac{(1-\pi)}{R} \tilde{b} - \tilde{q}\tilde{b}} \geq 0 \quad (\text{PC-I}, \xi)$$

Lemma 2. Suppose $y_d \leq \theta_L$, and the L-type defaults, that is, $\tilde{b} > \theta_L - y_d$ ($\chi = 0$). Therefore, investors make zero profits ($\xi > 0$), that is,

$$\tilde{q} = \frac{1 - \pi}{R}$$

Proof. Let $(\tilde{q}, \tilde{b})$ be a solution to (P). Assume, by contradiction, that investors make positive profits, that is, $\tilde{q} < \frac{1-\pi}{R}$. Since the H-type's payoff is strictly increasing in $\tilde{q}$, it must be the case that there exists $q' > \tilde{q}$ that increases the H-type's payoff. Since the payoff of

type L is also increasing in the price of the debt, and when $\tilde{q}$ PC-L is satisfied, we must have that q' also satisfies PC-L, which means that $\tilde{q}$ is not a solution to (P). $\square$

Proposition 4. *The risky pooling contract maximizes the payoff of the H-type among all contracts in which the L-type defaults.*

Proof.

$$(1 + \eta)u' \left(y + \frac{1 - \pi}{R} \times \tilde{b} \right) (1 - \pi) - R\beta u'(\theta_H - \tilde{b}) = R(v - \chi) \quad (5)$$

By looking for a solution in which the L-type, defaults, that is, $\tilde{b} \in (\theta_L - y_d, \theta_H - y_d)$, we have that $v = \chi = 0$. If the L-type's PC-I binds but PC-L is slack ($\eta = 0$), we have:

$$\tilde{b} = \frac{\theta_H - \frac{R\beta}{1 - \pi} \times y}{1 + \beta}$$

We now look for a parametrization in which $\eta = 0$. Notice that if $y_d = \theta_L$, PC-L is slack. By continuity of $u \left(y + \frac{1 - \pi}{R} \tilde{b} \right) + \beta u(y_d)$ in y_d , there must exist $\epsilon > 0$ such that PC-L holds with strict inequality for $y_d \in (\theta_L - \epsilon, \theta_L)$. Define:

$$\bar{\epsilon} \equiv \sup\{\epsilon : \epsilon > 0, \text{ such that PC-L holds with strict inequality for } y_d = \theta_L - \epsilon\}$$

By taking $y_d \equiv \theta_L - \bar{\epsilon}$ we have that $\eta = 0$. $\square$

A.3.2 Best Repayment Outcome

Proposition 5. *The safe pooling outcome maximizes the H-type's payoff in which: (i) lenders make zero profits; (ii) the sustainability constraint for the L-type binds; (iii) the L-type prefers to default rather than repay the H-type's contract.*

Proof. We start considering the problem in which both types repay and investors break even for each type. This case asks for two ICs for the L-type: (i) repays H-type's contract; (ii) defaults on the H-type's contract. Since both types repay, we also need a sustainability constraint for the H-type.

$$\max_{\{b_H, b_L, q_H, q_L\}} u(y + q_H b_H) + \beta u(\theta_H - b_H) \quad (S)$$

subject to

$$\begin{aligned}
b_H &\leq \theta_H - y_d & (\text{SUST-H}, \nu_H) \\
b_L &\leq \theta_L - y_d & (\text{SUST-L}, \nu_L) \\
u(y + q_L b_L) + \beta u(\theta_L - b_L) &\geq u(y + q_H b_H) + \beta u(y_d) & (\text{ICD-L}, \chi) \\
u(y + q_L b_L) + \beta u(\theta_L - b_L) &\geq u(y + q_H b_H) + \beta u(\theta_L - b_H) & (\text{ICR-L}, \lambda) \\
u(y + q_L b_L) + \beta u(\theta_L - b_L) &\geq v_L^a & (\text{PC-L}, \eta) \\
b_L \times \left[\frac{1}{R} - q_L \right] &\geq 0 & (\text{PCI-L}, \xi_L) \\
b_H \times \left[\frac{1}{R} - q_H \right] &\geq 0 & (\text{PCI-H}, \xi_H)
\end{aligned}$$

We look for a solution with $\xi_H, \xi_L > 0$, which implies $q_H = q_L = 1/R$. By the FOC with respect to (b_H) :

$$u' \left(y + \frac{b_L}{R} \right) - R\beta u'(\theta_L - b_L) = \frac{R\nu_L}{\chi + \lambda + \eta}$$

Consider first that case in which the sustainability constraint binds for the L-type: $\nu_L > 0$, that is, $b_L = \theta_L - y_d$. We can then rewrite the ICD-L as:

$$u \left(y + \frac{\theta_L - y_d}{R} \right) \geq u \left(y + \frac{b_H}{R} \right) \implies b_H \leq \theta_L - y_d$$

Since SUST-L binds we know that the L-type is indifferent between default and repayment of b_L . By the FOC with respect to b_H :

$$u'(y + b_H/R) - R\beta u'(\theta_H - b_H) = \chi u'(y + b_H/R) + \lambda[u'(y + b_H/R) - R\beta u'(\theta_L - b_H)]$$

If the relevant constraint is ICD-L (picking the H-type's contract and defaulting is better than picking the H-type's contract and repaying) it must be the case that:

$$u(y_d) \geq u(\theta_L - b_H)$$

That is:

$$b_H \geq \theta_L - y_d$$

□

A.4 Proof of Proposition 2

Lemma 3. *There exists $\pi_L(y_d)$ such that, for $\pi < \pi_L(y_d)$, the L-type sovereign prefers the risky pooling debt level.*

Proof. It's enough to check L-type's deviation to the *safe pooling*

$$\begin{aligned} y + \frac{1-\pi}{R} \left[\frac{\theta_H - \frac{\beta R y}{1-\pi}}{1+\beta} \right] &> y + \frac{\theta_L - y_d}{R} \\ \underbrace{1 - \left[\frac{(1+\beta)(\theta_L - y_d) + \beta R y}{\theta_H} \right]}_{\equiv \pi_L(y_d)} &> \pi \end{aligned}$$

□

We now check whether the H-type prefers the *risky pooling* over issuing $\theta_L - y_d$ at a risk-free price. Let the payoff of the *risky pooling* be defined as⁹:

$$\mathcal{F}(\pi) \equiv (1+\beta) \log \left(\frac{y + \frac{1-\pi}{R} \times \theta_H}{1+\beta} \right) + \beta \log \frac{\beta R}{1-\pi}$$

Lemma 4. $\mathcal{F}(\pi)$ has a minimum ρ' . Moreover, at its minimum:

$$\mathcal{F}(\rho') = \log(y) + \beta \log(\theta_H)$$

Proof. The first order condition of $\mathcal{F}$ is:

$$\mathcal{F}'(\rho') = (1+\beta) \times \frac{-\theta_H}{Ry + (1-\rho')\theta_H} + \frac{\beta}{1-\rho'} = 0$$

We then get a critical point $\rho' = 1 - \frac{\beta R y}{\theta_H}$. The second derivative of $\mathcal{F}$:

$$\mathcal{F}''(\pi) = -\frac{(1+\beta)\theta_H^2}{[Ry + (1-\pi)\theta_H]^2} + \frac{\beta}{(1-\pi)^2}$$

⁹Function $\mathcal{F}$ is obtained by plugging the equilibrium debt and prices of the *risky-pooling* equilibrium into the payoff of the high-type under repayment.

Evaluating the second derivative at the critical point ρ' we get:

$$\begin{aligned}\mathcal{F}''(\rho') &= \frac{-(1+\beta)\theta_H^2}{[Ry + (1-\rho')\theta_H]^2} + \frac{\beta}{(1-\rho')^2} \\ &= \frac{-1}{1+\beta} \times \left(\frac{\theta}{Ry}\right)^2 + \frac{1}{\beta} \times \left(\frac{\theta_H}{Ry}\right)^2 > 0\end{aligned}$$

□

Lemma 5. *The safe pooling payoff for the H-type is decreasing in y_d .*

Proof. We have already showed that when evaluated at ρ' , $\mathcal{F}$ attains its minimum and $\mathcal{F}(\rho') = \log(y) + \beta \log(\theta_H)$. We want to know under which conditions the *safe pooling* payoff of the H-type is decreasing in y_d , that is:

$$\begin{aligned}\frac{-1}{Ry + \theta_L - y_d} + \frac{\beta}{\theta_H - \theta_L + y_d} &< 0 \\ \frac{\beta}{\theta_H - \theta_L + y_d} &< \frac{1}{Ry + \theta_L - y_d} \\ \beta Ry + \beta \theta_L - \beta y_d &< \theta_H - \theta_L + y_d \\ \beta Ry + (1+\beta)\theta_L &< (1+\beta)y_d + \theta_H \\ \underbrace{\theta_L - y_d}_{b_{FD}^*(\theta_L)} &< \underbrace{\frac{\theta_H - \beta Ry}{1+\beta}}_{b_{FD}^*(\theta_H)}\end{aligned}$$

□

Lemma 6. *There exists $\pi_H(y_d)$ such that for $\pi < \pi_H(y_d)$, the H-type prefers the risky pooling debt level.*

Proof. Note that $\mathcal{F}$ is equal to the safe pooling payoff at its minimum, attained when $y_d = \theta_L$. Also, the safe pooling payoff is decreasing in y_d , which means that for $y_d < \theta_L$, the safe pooling payoff lies above the $\mathcal{F}$ at its minimum. By continuity of $\mathcal{F}$ we get the existence of $\pi_H(y_d)$ that makes the H-type indifferent between the safe and risky pooling outcomes. □

A.5 Comparison to Hidden Debt Models

A.5.1 Commitment vs. Monitoring

To make our model relatable to [Horn et al. \(2022\)](#), we provide a different interpretation of the disclosure policy. In the main text we have:

$$\phi(\pi) = \mathbb{1}_{\{V_{FD}(\pi) > V_{ND}(\pi)\}}$$

Suppose now that the sovereign has a set of available *messages* to be sent to the lenders: $M \equiv \{h, \ell\}$. We can express the disclosure policy as a conditional distribution over messages:

$$\begin{aligned}\phi &= \mathbb{P}(m = \ell | \theta = \theta_L) \\ 1 - \phi &= \mathbb{P}(m = h | \theta = \theta_L)\end{aligned}$$

Suppose that $\mathbb{P}(m = h | \theta = \theta_H) = 1$. The posterior belief of the lenders is then:

$$\mathbb{P}(\theta = \theta_L | m = h) = \frac{(1 - \phi)\pi}{(1 - \phi)\pi + 1 - \pi}$$

When $\phi = 0$, the sovereign *always sends the h-message*, making the posterior of the lender to be the same as the prior π (which is equivalent to no-disclosure in the main text). If $\phi = 1$, on the other hand, lenders are always informed (equivalent to FD in the main text). Suppose now we remove the commitment assumption, that is, the equivalent of Figure (2) is now:

Figure 7: Timing without the commitment assumption

t = 0	t = 1	t = 2
1) Government learns its type (θ_i)	1) Message is chosen $m \in \{h, \ell\}$ 2) Choice of debt 3) Price is determined 4) Consumption takes place	1) Default decision 2) Consumption

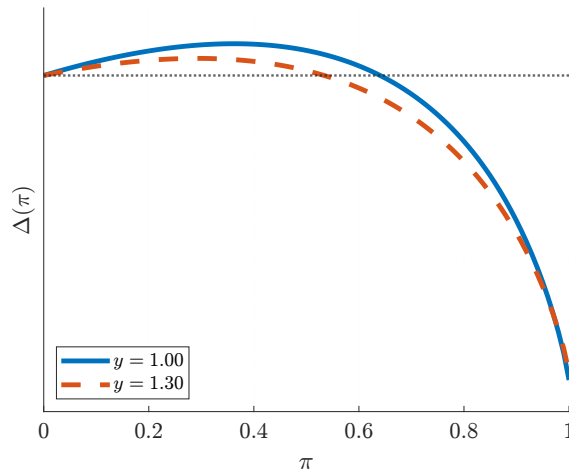
Under this timing, our model would feature a cheap talk equilibrium where both types would always send a h-message ([Crawford and Sobel, 1982](#)). This assumption micro-

finds the assumption from [Horn et al. \(2022\)](#) that governments cannot credibly communicate their state, transferring to the lenders the ability to monitor the sovereign and trigger revelations.

A.5.2 Comparative Statics (y).

[Horn et al. \(2022\)](#) shows that hidden sovereign debt tends to accumulate during boom years and is typically revealed during downturns. To relate this empirical pattern to our framework, we perform a comparative statics exercise with respect to the endowment parameter y . We interpret a low y as representing bad times, when resources are scarce. In [Horn et al. \(2022\)](#), lenders have stronger incentives to pay monitoring costs and uncover information during such periods. In contrast, in our setting, information revelation is a policy choice made by the government.

Figure 8: Difference between expected payoffs under ND and FD, for different beliefs.



Note: Parameters used: $\beta = 0.92$, $R = 1$, $y_d = (y_d^{\max} + y_d^{\min})/2$, $\theta_L = 1.25$, $\theta_H = 5$.

Figure (8) shows that for lower values of y , the range of parameters for which non-disclosure is preferred expands. The reason is that opacity enables low-type governments to borrow more; when income is low, this insurance motive dominates. Although this pattern may appear opposite to [Horn et al. \(2022\)](#), the difference arises because, in our model, the disclosure is endogenously chosen by the sovereign rather than driven by lenders incentives to monitor¹⁰.

¹⁰In a world where the government can credibly communicate its state to the market *and* lenders can monitor the sovereign, we would expect to see the monitoring fees being paid exactly in moments when the government wants to hide information from lenders.

A.6 Simple knife edge case

To simplify the analysis, we describe a set of conditions that express a limiting knife-edge case.

Assumption 2 (Limiting Case). *Suppose the following parametric restrictions hold:*

1. *The H-type is constrained (chooses the maximum amount of debt it could repay)*

$$y_d > \frac{\beta R}{1 + \beta} \left(\frac{\theta_H}{R} + \frac{y}{1 - \pi} \right)$$

2. *Low deadweight loss from default*

$$y_d \rightarrow \theta_L$$

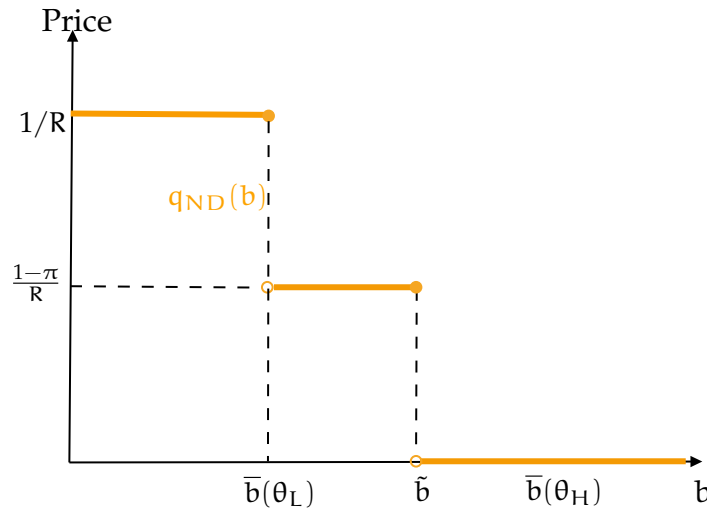
3. *H-type borrows at all*

$$\pi < 1 - \beta R$$

In the next proposition, we characterize the pooling equilibrium under Assumption (2).

Lemma 7. *Under Assumption (2), $\tilde{b} = \theta_H - y_d$ is the unique debt level sustained in a pooling equilibrium.*

Figure 9: Price schedule for the pooling eq.



Proof. Under the suggested belief system, the price schedule is represented in Figure (9).

The optimal choice of debt for a H-type sovereign that faces a price $\frac{1-\pi}{R}$ is:

$$\tilde{b} = \max \left\{ \min \left\{ \frac{\theta_H - R\beta \frac{y}{1-\pi}}{1+\beta}, \theta_H - y_d \right\} \right\}$$

In the limiting case $y_d \rightarrow \theta_L$ we thus have:

$$\tilde{b} = \min \left\{ \frac{\theta_H - R\beta \frac{y}{1-\pi}}{1+\beta}, \theta_H - \theta_L \right\}$$

Note that the H-type sovereign has positive marginal utility for an additional unit of debt, even when debt is at its upper limit. We can thus conclude that there is no profitable deviation for the H-type that would yield the same price, $\frac{1-\pi}{R}$. Since $y_d \rightarrow \theta_L$, no debt level can be issued at a risk-free price, which means that autarky is the only candidate for a profitable deviation for the H-type sovereign. For autarky to be preferred to the pooling equilibrium the following inequality must hold:

$$u \left(y + \frac{1-\pi}{R} \times (\theta_H - \theta_L) \right) - u(y) < \beta [u(\theta_H) - u(\theta_L)]$$

Which holds if $\pi > 1 - R\beta$, a contradiction. On the other hand, L-types' payoff is strictly increasing on debt, conditional on getting a positive price, which means they also will not deviate from $\theta_H - \theta_L$.

□

The above result is important for us to evaluate the sovereigns' disclosure policy as it makes sure that there is a *unique* payoff at $t = 0$ associated with not disclosing the aggregate state θ . We can now define such payoff as follows:

$$\begin{aligned} V_{ND}(y_d) \equiv & \pi \underbrace{\left[\log(y + q_{ND}(\tilde{b})\tilde{b}) + \beta \log(y_d) \right]}_{V_{ND}(\theta_L; y_d)} \\ & + (1 - \pi) \underbrace{\left[\log(y + q_{ND}(\tilde{b})\tilde{b}) + \beta \log(\theta_H - \tilde{b}) \right]}_{V_{ND}(\theta_H; y_d)} \end{aligned}$$

$$V_{ND}(y_d) = \log(y + q_{ND}(\tilde{b})\tilde{b}) + \beta[(1 - \pi) \log(\theta_H - \tilde{b}) + \pi \log(y_d)]$$

The disclosure policy picks the *sovereign-preferred equilibrium*, that is:

$$\phi^* = \arg \max_{\phi \in \{0,1\}} \phi \times V_{FD} + (1 - \phi) \times V_{ND}$$

Proposition 6. Under Assumption (2) $V_{ND} > V_{FD}$, that is, $\phi^* = 0$

Proof. Using the equilibrium outcomes derived, the expression for the payoffs under FD and ND reduces to:

$$\begin{aligned} V_{FD} &= \pi \times u(y) + (1 - \pi) \times u\left(y + \frac{\theta_H - \theta_L}{R}\right) + \beta u(\theta_L) \\ V_{ND} &= u\left(y + \frac{1 - \pi}{R} \times (\theta_H - \theta_L)\right) + \beta u(\theta_L) \end{aligned}$$

The result follows from the strict concavity of u . □

The above result shows that if there are no deadweight losses from defaulting at the L-state, not disclosing information to international lenders is preferred to disclosing it fully. By committing not to reveal θ , the sovereign can transfer resources from a realization of the H-state to the L-state by playing a pooling equilibrium. In this equilibrium, the L-type takes as much debt as the H-type, at a positive price, and defaults, while the H-type is getting a price lower than $1/R$ even though they would find it optimal to repay. Observe that our results do not critically depend on assuming a logarithmic utility function. Because the high-type sovereign is constrained, that is, it issues the maximum amount of debt that lenders are willing to purchase at a positive price, the sovereign's payoff is equalized across all realizations of the state variable θ . Due to the strict concavity of the utility function, the sovereign prefers this equalized payoff to the risky, lottery-like payoff that would result from a full disclosure policy.

Although restrictive, the limiting assumptions imposed above are useful to illustrate the mechanism that makes not disclosing preferred to the sovereign. In equilibrium, given that both the L and H-types are constrained, debt choices are $b \in \{\theta_L - y_d, \theta_H - y_d\}$. Considering the limiting case $y_d \rightarrow \theta_L$ makes sure that one of such choices is zero, providing a sharp characterization of the mechanism in our model by ruling out the issuance of risk-free debt. Bounding the probability of being an L-type by $1 - \beta R$ is slightly stronger than the $\beta R < 1$ assumption, and makes sure that the H-type prefers to take some positive debt.